

Lecture 26: Proteins, Molecules & Structural Biology

BINF 4002 — Machine Learning for Health

Matthew McDermott

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Dept. of Biomedical Informatics, Columbia University

Three lectures on biological & molecular data:

- L25: DNA, Genetics & Gene Regulation — genome-level biology + classical statistics
- L26: Proteins, Molecules & Structural Biology ← **today**
- L27: Modern Biological AI — deep learning across all levels

Same pattern as L25: biology interleaved with methods.

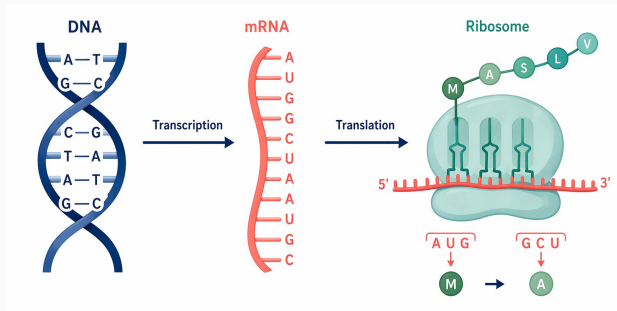
Today: Two molecular levels, each with their classical toolkit.

1. **Proteins:** sequences → alignment → conservation → variant prediction → structure
2. **Small molecules:** representations → fingerprints → QSAR → docking

These are the methods your DL models in L27 need to beat—and they're surprisingly hard to beat.

Protein Biology

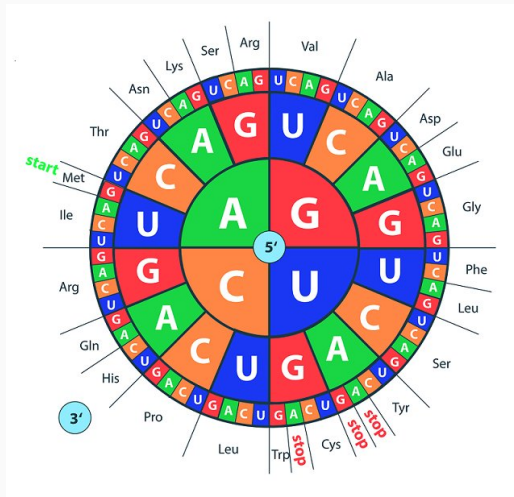
From DNA to Protein



DNA is transcribed to mRNA, then translated into protein by the ribosome. Each **codon** (3 nucleotides) specifies one amino acid. 64 codons → 20 amino acids + stop signals.

L25 covered the DNA side. Today: what happens after translation—the proteins and small molecules that do the work.

The Genetic Code



What Is a Protein?

A sequence of amino acids from a 20-letter alphabet.

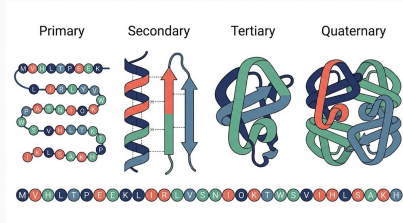
MVHLTPEEKSAVTALWGKVVNDEVGGEALGRLLVVYPWTQR...

Human hemoglobin β -chain (HBB), 147 residues.

The NLP analogy:

NLP concept	Protein analogue	Why it matters
Token	Amino acid (20 types)	Small vocabulary, no spaces
Word / n-gram	Motif (e.g., binding site)	Functional units
Phrase	Domain (~ 50 – 300 residues)	Modular building blocks
Grammar	3D fold constraints	Physics, not convention
Corpus	UniProt (>200 M seqs)	Massive unlabeled data

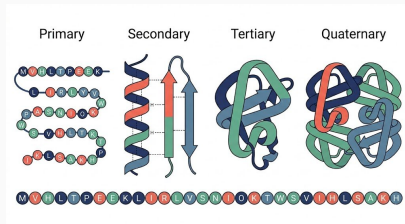
The Four Levels of Protein Structure



Sequence strongly constrains structure, which strongly constrains function.

(Environment, chaperones, PTMs, and binding partners also matter—but sequence is the starting point.)

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- Single amino acid change can dramatically alter behavior (sickle cell: E6V promotes HbS polymerization)
- Protein misfolding → disease (e.g., CFTR F508del in cystic fibrosis)
- Drugs bind specific 3D shapes → know the structure, design the drug

Experimental Structure Determination

Method	Characteristics
X-ray crystallography	Most PDB entries; requires crystals; often very high resolution
Cryo-EM	Growing fast; especially powerful for large complexes; no crystals needed
NMR	Solution-state ensembles; smaller systems; uniquely captures dynamics

Each has **biases**: what crystallizes $\neq$ what's most important biologically. Resolution varies by method and target. Cryo-EM has surged since $\sim$ 2015 (see Notebook Fig. 1).

The Protein Data Gap

The defining data challenge in protein biology:

- **Sequences:** >200 million in UniProt — massive, mostly unlabeled

This ratio screams “self-supervised pre-training” (L27). Before 2020, the sequence–structure gap was partially filled by classical methods. Understanding these is essential to understanding why AlphaFold 2 was a breakthrough.

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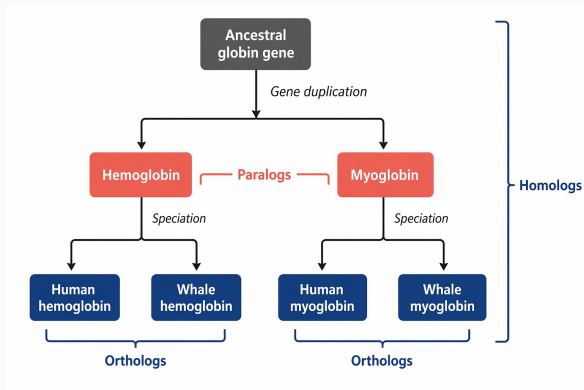
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- **Function labels:** ClinVar (clinical variants), DMS/MaveDB (experimental assays) — sparse but growing
- **Predicted structures:** ~200 million from AlphaFold DB — but predictions $\neq$ ground truth

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Evolutionary Relationships



Orthologs (speciation): usually conserve function.

Paralogs (gene duplication): may diverge in function.

All are **homologs**. Even distant homologs can share the same 3D fold—structure is more conserved than sequence.

Conserved positions = functional constraint.

Variable positions = often more tolerated, but can matter in specific contexts.

Classical Protein Analysis

Sequence Alignment: The Fundamental Operation

Pairwise alignment: match, mismatch, gap.

MVHLTP-EEK...

MLHLSPGEEK...

BLOSUM62: observed substitution frequencies. Intuition: “ $V \rightarrow L$ is cheap, $V \rightarrow P$ is expensive.”

Two modes:

- **Global** (Needleman-Wunsch): align entire sequences end-to-end
- **Local** (Smith-Waterman): find best-matching subsequence—used by BLAST, MMseqs2

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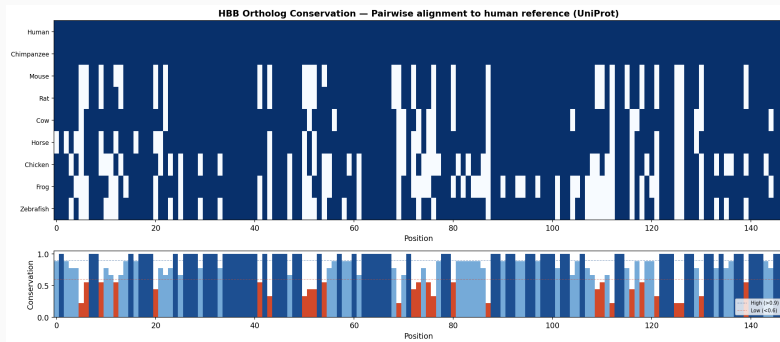
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Beyond pairwise: Profile HMMs (HHsearch, HMMER) model position-specific residue distributions—more sensitive for remote homologs; used in AlphaFold 2’s MSA pipeline.

Multiple Sequence Alignment (MSA)

Align N homologous sequences. The foundation of classical protein analysis.



[Notebook Fig. 2] Positional identity across 9 vertebrate HBB orthologs (UniProt). Dark = identical. Light = divergent. Everything downstream—conservation, coevolution, variant prediction, homology modeling—flows from the MSA. (This is a simplified conservation view; full MSA tools like Clustal Omega jointly optimize the alignment.)

Position-Specific Scoring Matrices:

- At each MSA position, compute probability of each amino acid
- Encodes which substitutions are tolerated *at that position*
- Used for: variant scoring, PSI-BLAST
- **Limitation:** positions scored independently—ignores context

Coevolution & Contact Prediction:

- Positions that mutate *together* → often in physical contact
- Direct Coupling Analysis (DCA)
- **AlphaFold 2 explicitly uses** MSA coevolution; protein LMs may recover related signals implicitly

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Intuition

If positions 12 and 85 always change together across species, they are probably in 3D contact. The MSA contains implicit structural information.

The Variant Effect Problem

A mutation changes the DNA sequence. What happens to the protein?

Variant type	What happens	Likely impact
Synonymous	Different codon, same amino acid	Usually neutral (but not always)
Missense	Different amino acid at one position	Ranges from benign to lethal
Nonsense	Premature stop codon	Usually loss of function
Frameshift	Insertion/deletion shifts reading frame	Usually loss of function

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The key question for clinical genetics: given a missense variant, is it benign or pathogenic? This is the *variant effect prediction* problem—and it is where conservation from the MSA becomes directly useful.

Classical Variant Effect Prediction

Pre-DL methods — many still used in clinical genetics:

Method	Approach	Scope
SIFT	MSA conservation → continuous score	Missense variants
PolyPhen-2	Sequence + structural features	Missense variants
CADD	Integrates > 60 annotations	All variant types
REVEL	Ensemble of missense-specific scores	Missense variants

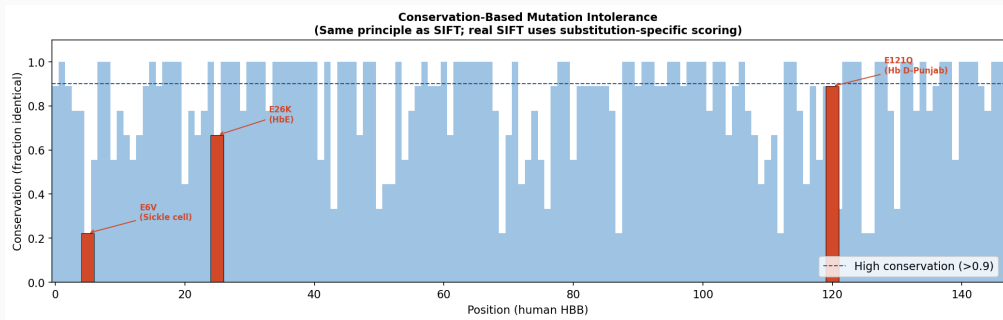
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A shared signal is **evolutionary conservation**, though these methods differ substantially in what additional features they incorporate. In current practice, integrative scores (REVEL, CADD) and DL methods (AlphaMissense) are increasingly preferred.

Variant Effect Prediction: Conservation in Action

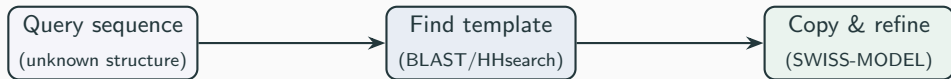


[Notebook Fig. 3] Conservation across HBB orthologs. Highly conserved positions (dark blue) are intolerant of substitution—mutations there are more likely damaging. This is the principle underlying SIFT; real SIFT uses substitution-specific scoring from the full MSA.

Homology Modeling: Predicting Structure Before AlphaFold

Task: Given a protein sequence with unknown structure, predict its 3D fold.

Classical approach: If a homolog with roughly $>30\%$ sequence identity has a solved structure, use it as a template.



Limitation: Fails for novel folds (no template). Before AlphaFold 2 (2020), $\sim 30\%$ of protein families had no structural template.

The Protein Analysis Toolkit (Summary)

MSA $\rightarrow$ conservation / coevolution $\rightarrow$ variant prediction / homology modeling.

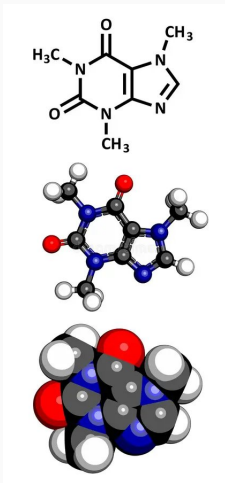
Everything flows from the MSA.

If proteins are the machinery of the cell,
how do we design molecules to interact with them?

Small molecules are one answer—
biologics, gene therapies, and antibodies are others.

Small Molecules

Molecular Representations: Three Different Ways



Caffeine shown three ways:

2D Skeletal (top)

The chemist's standard notation.

Ball-and-stick (middle)

Atoms and bonds in 3D. Shows geometry.

Space-filling (bottom)

Van der Waals radii. Shows molecular shape.

SMILES string:

Cn1c(=O)c2c(ncn2C)n(C)c1=O

Representation choice → model family:

SMILES → sequence models

Graph → GNNs

3D coordinates → equivariant networks

SMILES: Molecules as Strings

SMILES (Simplified Molecular-Input Line-Entry System):

- Atoms → element symbols: C, N, O, S
- Branches → parentheses: C(=O)O
- Rings → digit pairs: c1ccccc1 (benzene)
- Chirality → @, @@ (but often omitted)

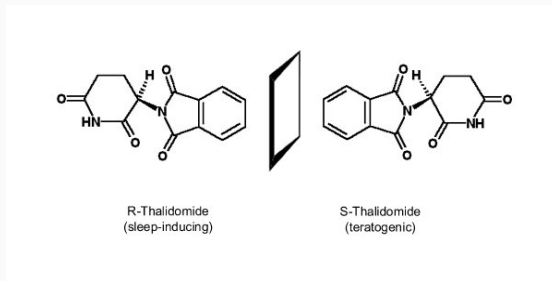
Aspirin: CC(=O)Oc1ccccc1C(=O)O

Representation → Model Family

SMILES (1D text) → sequence models. Graph (2D: atoms = nodes, bonds = edges) → GNNs. 3D coordinates → equivariant networks (L27).

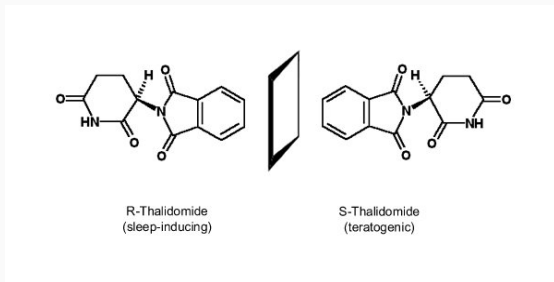
Limitations: Not unique (multiple valid SMILES per molecule). Chirality notation exists but is often omitted. No explicit 3D information.

Chirality: Why 3D Shape Matters



Enantiomers: non-superimposable mirror images at a chiral center. Identical 2D structure, different 3D handedness → different biological effects.

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Thalidomide: R-enantiomer is sleep-inducing; S-enantiomer is teratogenic. But they interconvert in vivo—you cannot simply administer “the safe one.” This is why 2D representations (SMILES, fingerprints) lose critical information for drug design.

Morgan/ECFP fingerprints: hash local substructures into binary vectors.

1. Walk outward from each atom (radius 2–3 bonds)
2. Hash each substructure → set bit in vector (e.g., 2048 bits)
3. **Tanimoto similarity:** $\frac{|A \cap B|}{|A \cup B|}$ for comparing molecules

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Activity cliffs: molecule pairs with high Tanimoto similarity but dramatically different potency.
Fingerprints miss subtle structural changes that flip biological activity.

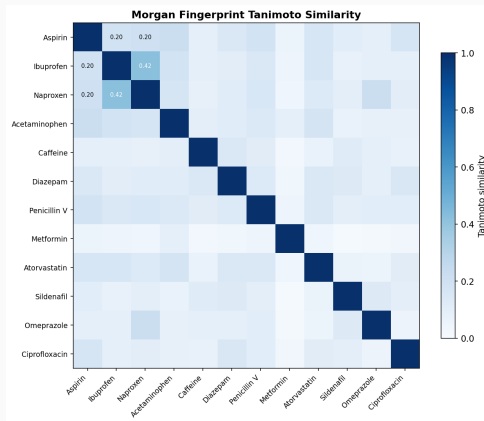
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Still competitive! FP + RF/XGBoost often beats GNNs on small datasets.

Fingerprint Similarity in Practice



[Notebook Fig. 4] Tanimoto similarity matrix for 12 well-known drugs. Structurally related compounds (e.g., NSAIDs: aspirin, ibuprofen, naproxen) cluster together; unrelated drugs show low similarity.

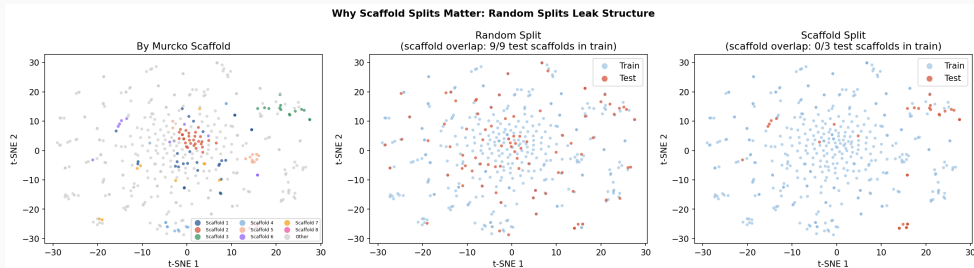
QSAR (Quantitative Structure–Activity Relationships): fingerprint $\rightarrow$ RF/SVM/XGBoost $\rightarrow$ property prediction (toxic? soluble? active?).

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Why evaluation matters: Random train/test splits *leak scaffolds*—molecules with the same core backbone appear in both sets, inflating apparent performance.

Scaffold splits: Entire molecular scaffolds are held out for testing. This tests *extrapolation* to new chemistry, not interpolation within known chemotypes. Analogous to patient-level splits in clinical ML and cross-ancestry PRS evaluation from L25.

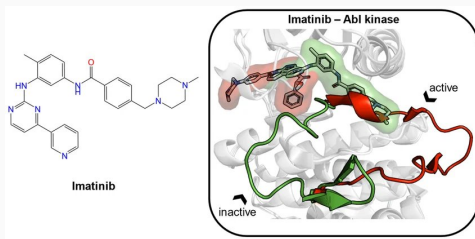
Scaffold Splits: Why Random Splits Fail



[Notebook Fig. 5] Left: molecules colored by Murcko scaffold. Middle: random split—scaffolds leak across train/test. Right: scaffold split—test scaffolds are entirely unseen during training.

Molecular Docking

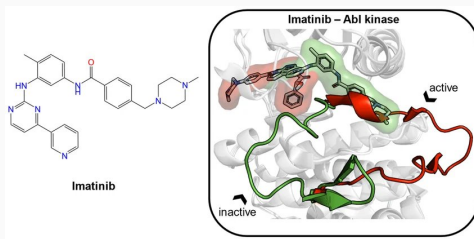
This is where proteins and small molecules meet: given a protein structure and a candidate drug, predict how and how well they bind.



Imatinib (Gleevec) bound to Abl kinase (PDB: 1IEP). Left: 2D structure. Right: drug in the binding pocket—shape complementarity in action.

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Virtual screening: Dock millions of candidates, test top hits. Scoring: force-field-based, empirical, or knowledge-based. Useful for filtering and hypothesis generation, but raw docking scores $\neq$ true binding affinity.

Integration

The Drug Discovery Pipeline



Where classical methods fit at each step:

- **Target:** GWAS → eQTL → which protein to target

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- **Lead optimization:** QSAR, fingerprint similarity

Classical Methods: Baselines to Beat or Build Upon

Task	Classical method	Strength
Variant effect	SIFT, PolyPhen, REVEL, CADD	Widely used, strong baselines
Structure prediction	Homology modeling	Works if template exists
Molecular similarity	Morgan FP + Tanimoto	Fast, no training needed
Property prediction	FP + RF	Often competitive on small data
Binding prediction	Docking	General-purpose baseline

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Note: the MSA is not only a classical method—it remains a *critical input* to the best DL models. AlphaFold 2's Evoformer explicitly consumes MSA columns. Classical and DL are not always in competition; often the best systems integrate both.

Key Takeaways

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- **Activity cliffs** show that structural similarity $\neq$ functional similarity
- **Scaffold split** is the molecular patient-level split — evaluation $>$ model choice
- **Classical + DL**: the best systems integrate classical tools (MSAs, fingerprints) with learned representations

Acronyms and Tools Introduced Today

Acronym	Full Name
BLAST	Basic Local Alignment Search Tool
PSI-BLAST	Position-Specific Iterated BLAST
HMM	Hidden Markov Model
PSSM	Position-Specific Scoring Matrix
DCA	Direct Coupling Analysis
SIFT	Sorting Intolerant From Tolerant
PolyPhen	Polymorphism Phenotyping
CADD	Combined Annotation Dependent Depletion
REVEL	Rare Exome Variant Ensemble Learner
SMILES	Simplified Molecular-Input Line-Entry System
ECFP / Morgan	Extended-Connectivity Fingerprint
QSAR	Quantitative Structure–Activity Relationship

Questions?